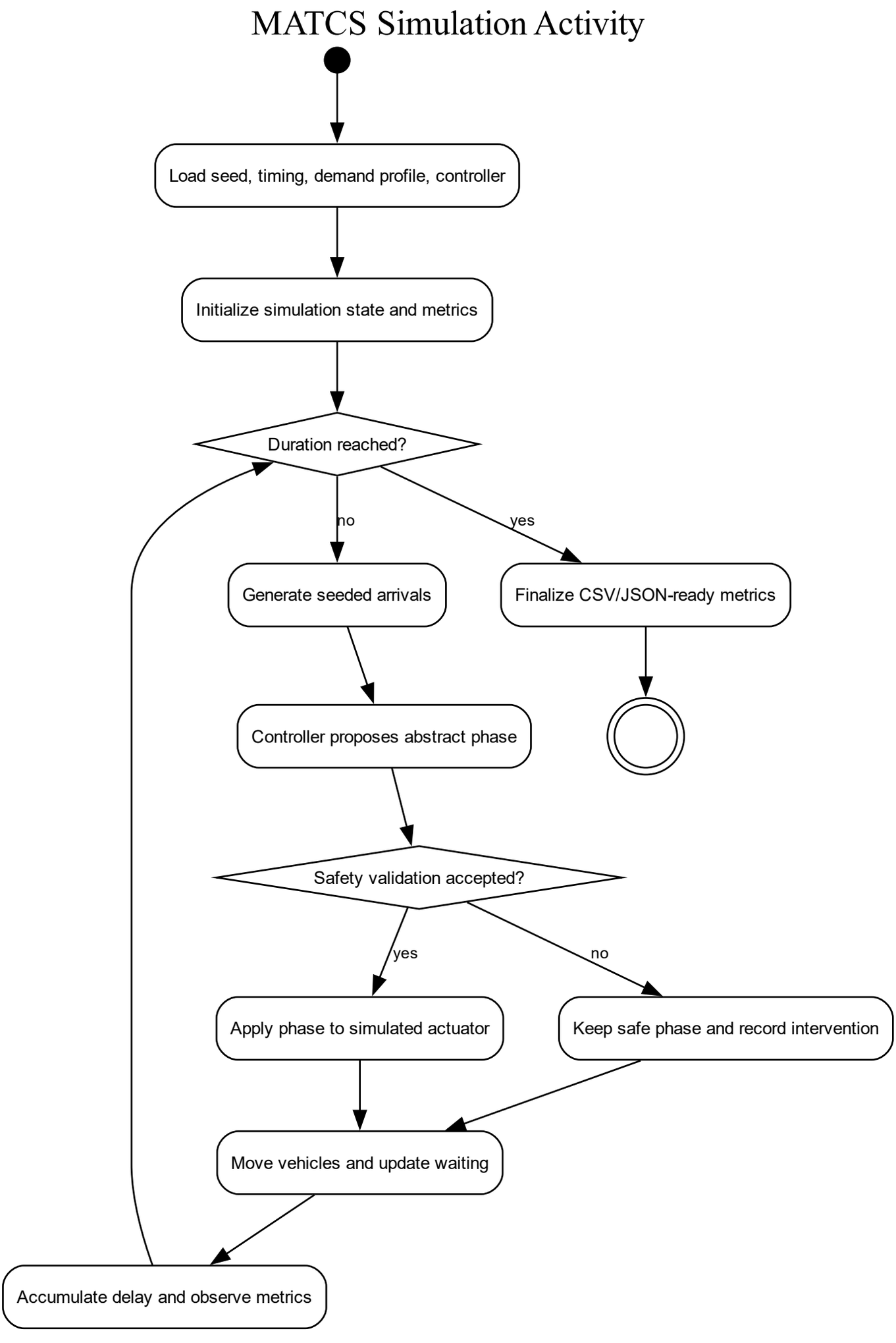
MATCS Simulation Activity
Load seed, timing, demand profile, controller
Initialize simulation state and metrics
Duration reached?
no
yes
Generate seeded arrivals
Finalize CSV/JSON-ready metrics
Controller proposes abstract phase
Safety validation accepted?
yes
no
Apply phase to simulated actuator
Keep safe phase and record intervention
Move vehicles and update waiting
Accumulate delay and observe metrics